Prelab 4

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What does a root locus plot depict?

Ans. to the Q. No. 1(a)

A root locus plot depicts how the closed-loop poles of a system move through the complex s-plane as the gain K varies from 0 to ∞. Each point on the locus is a location where a closed-loop pole exists for some value of K. It gives a visual picture of how a system's stability and dynamic response change with gain.

What must be done to a transfer function before its root locus can be graphed?

Ans. to the Q. No. 1(b)

The open-loop transfer function must be written in root locus form: $1 + K \cdot G(s)H(s) = 0$

Therefore, the following 3 steps need to be done:

i. Form the characteristic equation of the closed-loop system.

ii. Factor out K so the equation is in the form $1 + K \cdot G(s) \cdot H(s) = 0$.

iii. Identify the open-loop poles (denominator roots) and open-loop zeros (numerator roots) of $G(s)H(s)$.

What is the significance of the gain K?

Ans. to the Q. No. 1(c)

K is the variable parameter that drives the root locus. Its significance includes:

At K=0, the closed-loop poles begin at the open-loop poles.
As K→∞, the closed-loop poles travel towards the open-loop zeros (or to infinity along asymptotes). K directly controls the transient response; rise time, overshoot, settling time, and damping.
It determines whether the system is stable, marginally stable, or unstable (poles in left/right half plane). A specific value of K can be selected to meet a desired performance specification.

How can a root locus plot be used to design a controller?

Ans. to the Q. No. 1(d)

By overlaying design specification regions on the s-plane (i.e., desired damping ratio ζ, natural frequency ω_n, or settling time constraints), a designer can:

i. Identify a desired closed-loop pole location on the s-plane.

ii. Find the gain K that places a pole at that location using the magnitude condition.

iii. If no point on the existing locus meets the specifications, add a compensator (lead, lag, PD, PI, or PID), which adds poles and zeros to reshape the locus so it passes through the desired region.

iv. Verify stability margins and check that all closed-loop poles result in acceptable behaviour.

Rules for completing the root locus plot using pencil and paper

Ans. to the Q. No. 1(e)

The five rules are taken from Nise Control Systems Engineering textbook Section 8.4:

i. Number of branches: The number of branches equals the number of closed-loop poles.

ii. Symmetry: The root locus is symmetrical about the real axis.

iii Real-axis segments: For K>0, the root locus exists to the left of an odd number of real-axis, finite open-loop poles and/or finite open-loop zeros.

iv. Starting and ending points: The locus begins (K=0) at the finite and infinite poles of G(s)H(s) and ends (K→∞) at the finite and infinite zeros of G(s)H(s).

V. Behaviour at infinity: The locus approaches straight-line asymptotes as it goes to infinity, defined by:

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\#\text{finite poles} - \#\text{finite zeros}}$$

$$\theta_a = \frac{(2k+1)\pi}{\#\text{finite poles} - \#\text{finite zeros}}, \quad k = 0, \pm1, \pm2, \ldots$$

Ans. to Q. No. 2(a)

$$G(s) = \frac{1}{(s+5)(s+9)}$$

first calculate the asymptotes.

$$\text{Using} \quad \sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\#\text{finite poles} - \#\text{finite zeros}}$$

$$= \frac{(-5-9) - 0}{2-0}$$

$$= \frac{-14}{2}$$

$$= -7$$

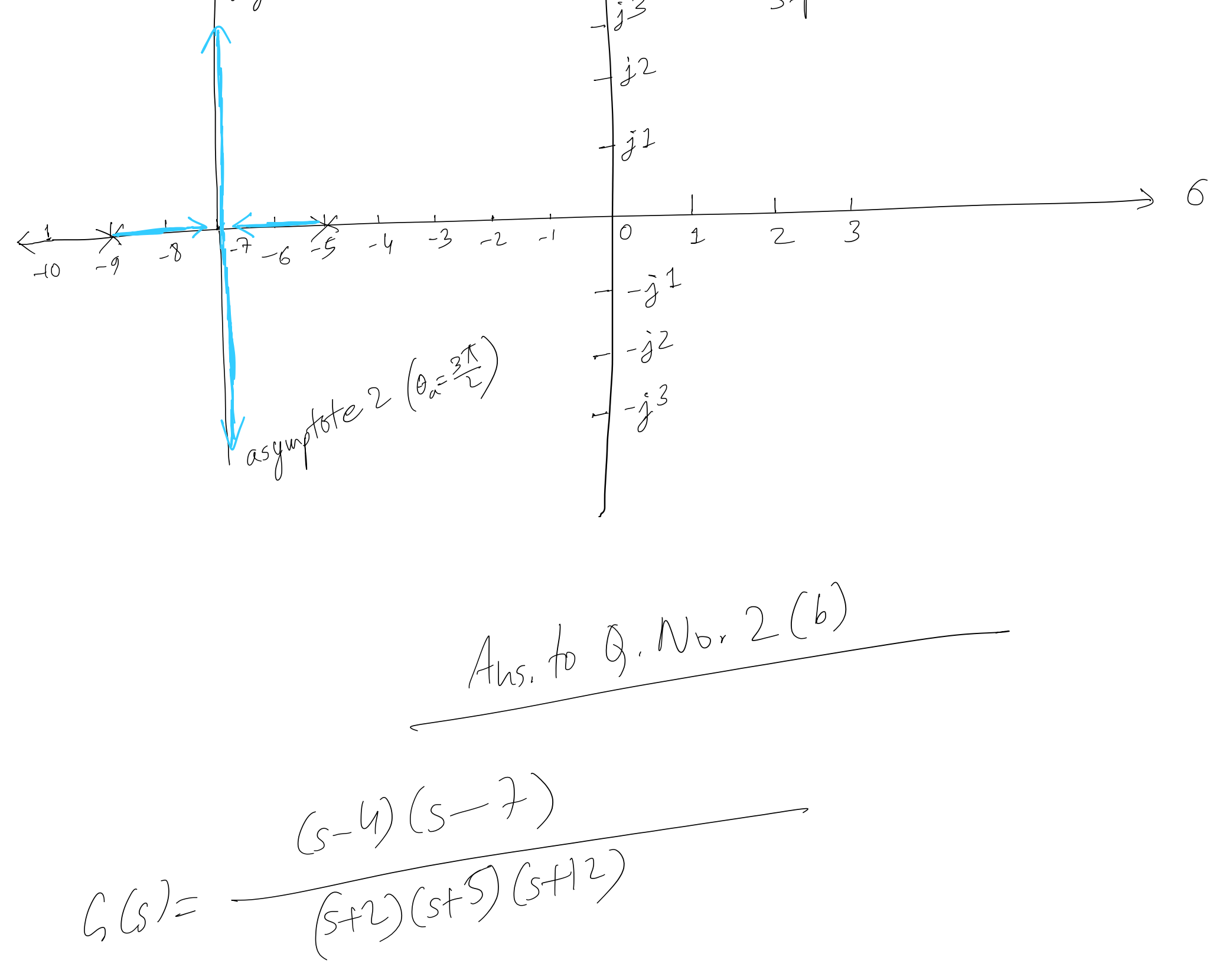
The angles of lines that intersect at -7

$$\theta_a = \frac{(2k+1)\pi}{\#\text{finite poles} - \#\text{finite zeros}}$$

if $k=0, \quad \theta_a = \frac{\pi}{2}$

if $k=1, \quad \theta_a = \frac{3\pi}{2}$

The number of lines obtained equals the difference between the number of finite poles and the number of finite zeros, which is 2. (in question 2(a)) Rule 4 states that the locus begins at the open-loop poles and ends at the open-loop zeros. Since there are more open-loop poles than open-loop zeros in $G(s) = \frac{1}{(s+5)(s+9)}$, there must be zeros at infinity. The asymptotes tell us how we get to these zeros at infinity.



— root locus
jω
asymptote 1 ($\theta_a = \frac{\pi}{2}$)
s-plane
j3
j2
j1
-10
-9
-8
-7
-6
-5
-4
-3
-2
-1
0
1
2
3
σ
-j1
-j2
-j3
asymptote 2 ($\theta_a = \frac{3\pi}{2}$)

Ans. to Q. No. 2(b)

$$G(s) = \frac{(s-4)(s-7)}{(s+2)(s+5)(s+12)}$$

calculate the asymptotes:

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\#\text{finite poles} - \#\text{finite zeros}}$$

$$= \frac{(-2-5-12) - (4+7)}{3-2}$$

$$= \frac{-19-11}{1}$$

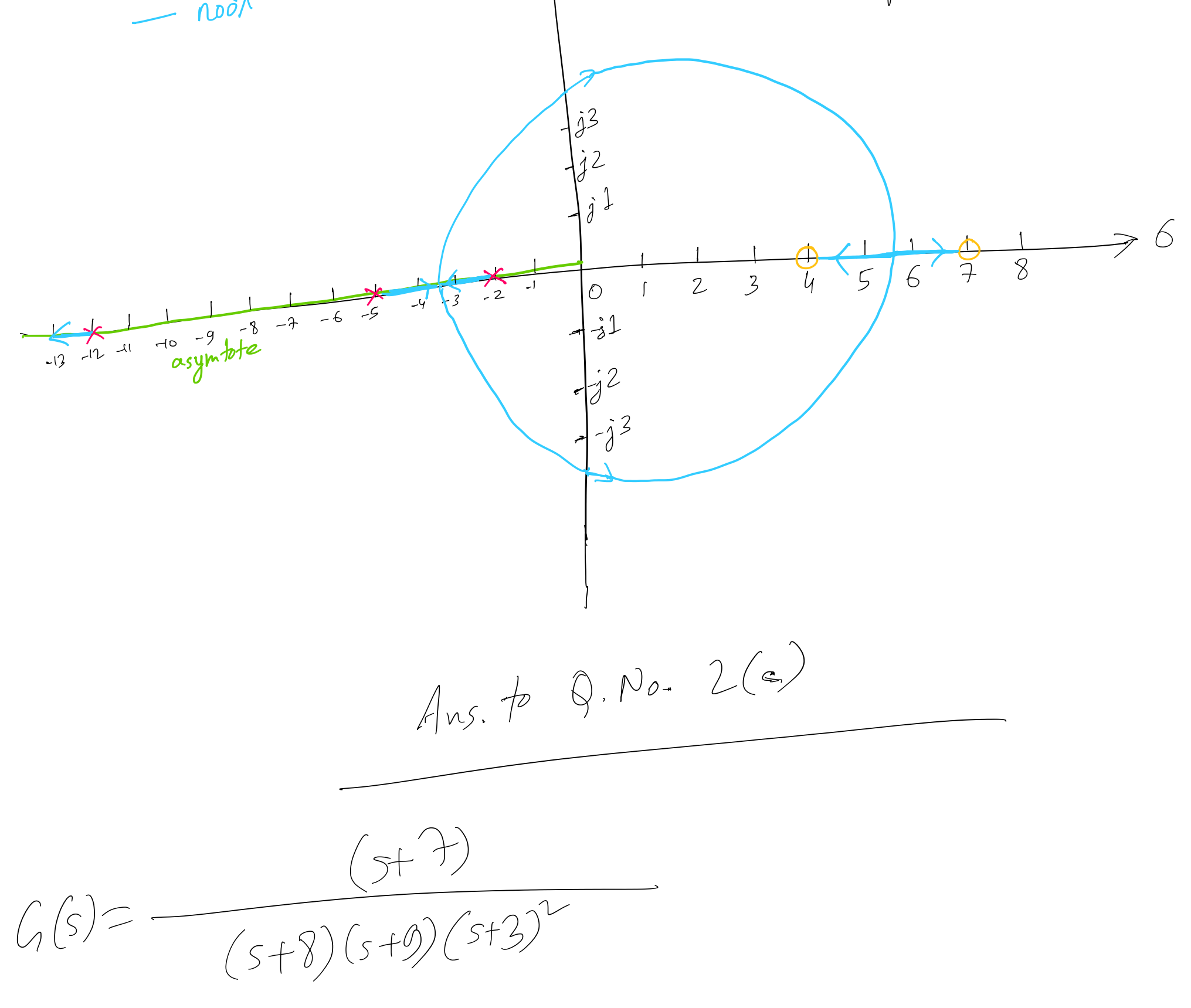
$$= -30$$

The angles of lines that intersect at -30,

$$\theta_a = \frac{(2k+1)\pi}{\#\text{finite poles} - \#\text{finite zeros}}$$

if $k=0, \quad \theta_a = \frac{\pi}{1} = \pi$

The number of lines obtained equals the difference between the number of finite poles and the number of finite zeros, which is 1 for question 2(b). Rule 4 states that the locus begins at open-loop poles and ends at open-loop zeros. Since there are more open-loop poles than open-loop zeros, there must be zeros at infinity.



Ans. to Q. No. 2(c)

$$G(s) = \frac{(s+7)}{(s+8)(s+9)(s+3)^2}$$

Calculate the asysmptotes:

$$\sigma_a = \frac{\sum \text{finite poles} - \sum \text{finite zeros}}{\#\text{finite poles} - \#\text{finite zeros}}$$

$$= \frac{(-8-9-3-3) - (-7)}{4-1}$$

$$= \frac{-16}{3}$$

The angles of the lines that intersect at $\frac{-16}{3}$,

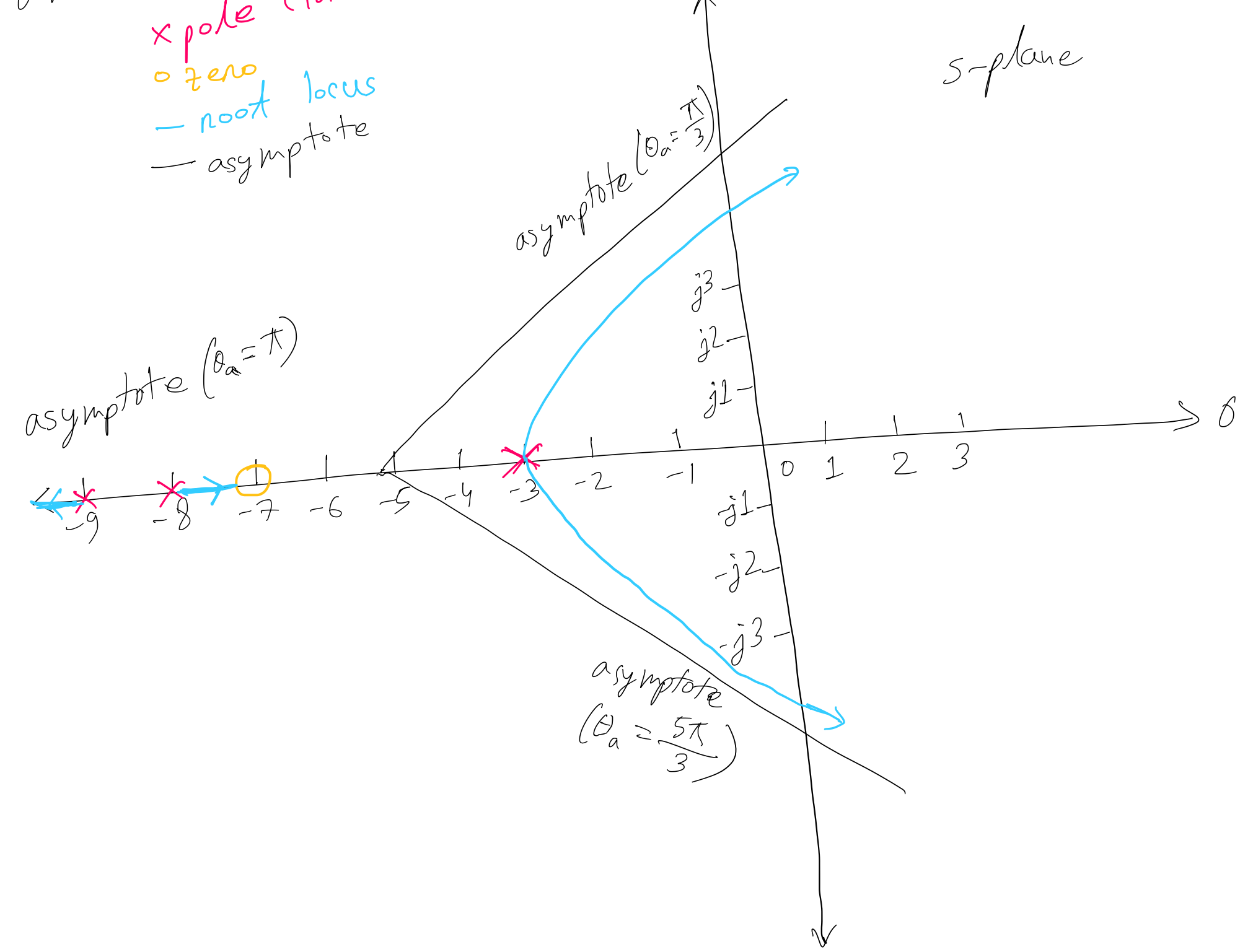
$$\theta_a = \frac{(2k+1)\pi}{\#\text{finite poles} - \#\text{finite zeros}}$$

if $k=0, \quad \theta_a = \frac{\pi}{3}$

if $k=1, \quad \theta_a = \frac{3\pi}{3} = \pi$

if $k=2, \quad \theta_a = \frac{5\pi}{3}$

The number of lines obtained equals the difference between the number of finite poles and the number of finite zeros, which is, 4 - 1 = 3 for question 2(c).



× pole (total 4 poles)
o zero
— root locus
— asymptote
jω
s-plane
asymptote ($\theta_a = \frac{\pi}{3}$)
asymptote ($\theta_a = \pi$)
j3
j2
j1
-9
-8
-7
-6
-5
-4
-3
-2
-1
0
1
2
3
σ
-j1
-j2
-j3
asymptote ($\theta_a = \frac{5\pi}{3}$)